

THA PHRA AGROMET, Thailand

WMO: 483840

Lat: 16.335N

Lon: 102.827E

Elev: 166

StdP: 99.35

Time Zone: 7.00 (E07)

Period: 08-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 13.7	(c) 15.5	(d) 9.2	(e) 7.4	(f) 18.0	(g) 11.0	(h) 8.3	(i) 18.4	(j) 4.2	(k) 25.0	(l) 3.7	(m) 24.1	(n) 0.3	(o) N/A	(p) 0.298

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
4	10.1	38.5	25.2	37.2	25.5	35.9	25.6	28.0	33.1	27.7	32.6	27.4	32.1	1.5	N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
26.9	23.0	30.1	26.6	22.6	29.7	26.2	22.1	29.3	91.1	33.1	89.4	32.8	88.1	32.4	29.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
Min		Max		Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
3.7	3.2	2.6	DB	10.6	39.5	1.8	1.6	9.3	40.7	8.2	41.6	7.1	42.5	5.8	43.6	
			WB	9.7	29.0	2.0	0.6	8.3	29.4	7.1	29.7	6.0	30.0	4.6	30.4	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	27.7	23.8	26.1	28.8	30.5	29.9	29.6	28.7	28.3	28.2	27.7	26.4	24.0
	DBStd	2.96	2.68	2.93	2.93	2.49	1.78	1.28	1.57	1.21	1.14	1.31	1.94	2.68
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD18.3	2	2	0	0	0	0	0	0	0	0	0	0	1
	CDD10.0	6447	427	451	582	615	616	587	579	569	545	549	493	435
	CDD18.3	3408	170	217	324	365	358	337	321	311	295	291	243	177
	CDH23.3	36982	1399	2176	3906	4718	4331	4101	3624	3289	3003	2804	2168	1463
	CDH26.7	16332	522	1038	2074	2636	2091	1854	1432	1153	1033	1055	883	561
Wind	WSAvg	0.5	0.5	0.5	0.6	0.6	0.5	0.6	0.6	0.6	0.4	0.4	0.5	0.6
Precipitation	PrecAvg	1237	8	18	42	93	162	151	178	220	243	100	19	5
	PrecMax	1741	63	69	124	287	295	318	327	405	452	192	106	37
	PrecMin	903	0	0	0	19	50	54	29	83	76	8	0	0
	PrecStd	203	15	20	37	66	65	58	67	73	90	56	28	9
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	34.0	37.3	39.4	40.5	38.5	36.3	36.9	34.6	34.2	34.5	34.8	34.5
		MCWB	23.2	24.0	23.5	25.6	26.6	26.7	26.6	26.6	27.3	26.4	25.8	23.9
	2%	DB	32.6	35.7	38.0	39.0	36.8	35.1	34.7	33.3	33.3	33.4	33.5	33.0
		MCWB	23.1	23.8	24.2	25.7	26.6	26.7	26.6	26.7	27.1	26.1	25.3	23.8
	5%	DB	31.2	34.1	36.6	37.5	35.4	34.0	33.3	32.3	32.3	32.5	32.5	31.5
		MCWB	22.8	23.6	24.6	25.9	26.8	26.7	26.5	26.7	26.9	26.0	25.0	23.2
	10%	DB	29.8	32.5	35.1	36.0	34.0	33.0	32.0	31.4	31.4	31.6	31.3	29.9
		MCWB	22.3	23.5	24.7	25.9	27.0	26.6	26.2	26.5	26.7	25.8	24.5	22.4
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	25.5	26.3	27.1	28.0	28.6	28.4	27.7	28.0	28.2	27.7	27.0	25.4
		MCDB	30.6	33.4	34.9	35.4	34.0	33.2	32.7	31.7	32.3	32.2	32.3	32.0
	2%	WB	24.6	25.4	26.4	27.4	28.0	27.8	27.2	27.5	27.6	27.1	26.2	24.5
		MCDB	29.7	32.4	33.9	34.7	33.7	32.4	31.9	31.2	31.6	31.5	31.6	31.0
	5%	WB	23.8	24.8	25.8	27.0	27.6	27.4	26.9	27.1	27.3	26.6	25.8	23.8
		MCDB	28.6	31.4	33.0	34.0	32.9	31.9	31.3	30.9	31.0	30.7	30.9	29.9
	10%	WB	23.1	24.2	25.4	26.5	27.2	27.0	26.5	26.7	27.0	26.2	25.2	23.1
		MCDB	28.0	30.5	32.1	33.2	32.2	31.5	30.7	30.2	30.5	29.9	29.3	28.4
Mean Daily Temperature Range	5% DB	MDBR	12.2	12.6	11.4	10.1	8.3	7.2	6.5	6.4	6.7	7.9	9.9	11.9
		MCDBR	13.3	14.0	13.3	11.4	9.9	8.5	8.5	7.7	8.0	9.1	10.9	12.7
		MCWBR	4.7	4.0	3.3	2.8	2.6	2.4	2.5	2.6	2.9	3.1	3.5	4.4
	5% WB	MCDBR	11.0	11.6	11.2	10.3	8.9	7.9	7.4	7.0	7.3	8.0	9.6	11.6
MCWBR		4.4	3.8	3.5	3.1	3.1	2.7	2.5	2.9	3.0	3.1	3.3	4.3	
Clear-Sky Solar Irradiance	taub		0.479	0.589	0.656	0.615	0.510	0.461	0.450	0.459	0.503	0.539	0.477	0.471
	taud		2.075	1.807	1.694	1.830	2.150	2.307	2.339	2.301	2.164	2.027	2.170	2.120
	Ebn at Noon		805	733	700	731	801	832	843	843	809	766	802	799
	Edh at Noon		158	216	248	217	155	131	128	134	153	171	142	147
All-Sky Solar Radiation	RadAvg		4.88	5.19	5.30	5.71	5.52	5.18	4.83	4.71	4.62	4.93	4.90	4.74
	RadStd		0.29	0.26	0.26	0.23	0.29	0.27	0.33	0.28	0.31	0.35	0.18	0.12

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (24 neighbors)	+0.30	N/A	N/A	N/A	+0.17	+0.14	N/A	N/A	+120	+122

Nomenclature: See separate page